# Stop Memorizing.
# Start Understanding Why.

Elaborative Interrogation · Woloshyn, Pressley & Schneider

> **"I memorized the answer.**
> **But when the question was phrased differently — gone."**
>
> — Rote facts are brittle. Understanding is robust.

There is a profound difference between knowing that something is true and knowing why it is true. A bare fact — memorized in isolation — is easy to forget and hard to apply when a question is worded unexpectedly. The same fact anchored to a causal explanation is nearly impossible to forget, because your brain can reconstruct it even when direct recall fails.

## The Research

**Woloshyn, Pressley, and Schneider (1992)** demonstrated that students who asked "why does this make sense?" as they studied significantly outperformed students who used repetition alone — including on delayed tests. **Pressley and colleagues (1992)** extended this across science, history, and medical content: when learners generated their own explanations, memory was more durable, transfer to novel problems was stronger, and confusable items were better discriminated.

> **The key finding:** Generating your own explanation is significantly more powerful than reading a provided explanation — even if the provided one is more accurate. The act of generation drives encoding, not the correctness of the result (Pressley et al., 1992).

The mechanism is elaborative encoding: each "why" connects the new fact to existing knowledge, creating multiple retrieval pathways. A fact connected to five other concepts is five times harder to forget than a fact standing alone.

## What This Looks Like

| Rote Fact | Elaborated Understanding |
|---|---|
| **Heart rate increases during exercise** | "Muscles demand more oxygen → heart pumps faster to deliver it. The same mechanism explains why anxiety raises heart rate — sympathetic activation signals the same 'urgent demand' response." |
| **Antibiotics don't work on viruses** | "Antibiotics target bacterial cell walls — viruses have no cell wall. Giving antibiotics for a virus is like using a key on a lock that doesn't exist." |

| Rote Fact | Elaborated Understanding |
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| **Spaced repetition improves retention** | "Memory decays over time, but each retrieval resets the clock and slows the next decay. Spacing exploits this: review just before forgetting, rebuild the trace stronger each time." |
| **Growth mindset improves performance** | "When students believe ability is developable, setbacks signal 'I need a different strategy' rather than 'I'm not smart enough.' The belief changes the behavioral response to failure." |

## The Protocol: Three Questions

For any fact you're trying to learn, run it through three questions before moving on:

**1. Why does this make sense?**
Don't accept the fact — demand a mechanism. "Because the book says so" is not an answer. Keep asking until you hit a principle your brain already understands.

**2. How does this connect to what I already know?**
Look for bridges to prior knowledge — from earlier in the course, other subjects, lived experience. Every connection is a new retrieval pathway.

**3. What would the opposite look like — and why?**
Contrast forces discrimination. If you understand why A is true, explaining why not-A clarifies A at a deeper level — and makes them harder to confuse on an exam.

**"Memory is a web, not a list. Every 'why' you generate is a new strand — and strands don't break the way isolated facts do." — adapted from Pressley et al. (1992)**

## REFERENCES

Woloshyn, V. E., Pressley, M., & Schneider, W. (1992). Elaborative-interrogation and prior-knowledge effects on learning of facts. Journal of Educational Psychology, 84(1), 115–124.

Pressley, M., Wood, E., Woloshyn, V. E., Martin, V., King, A., & Menke, D. (1992). Encouraging mindful use of prior knowledge: Attempting to construct explanatory answers facilitates learning. Educational Psychologist, 27(1), 91–109.

Willoughby, T., & Wood, E. (1994). Elaborative interrogation examined at encoding and retrieval. Learning and Instruction, 4(2), 139–149.

McDaniel, M. A., & Donnelly, C. M. (1996). Learning with analogy and elaborative interrogation. Journal of Educational Psychology, 88(3), 508–519.